

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON
ACADEMIC MANAGEMENT SYSTEM

P R O J E C T R E P O R T

Submitted by

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JAVA ASSIGNMENT

AIM:

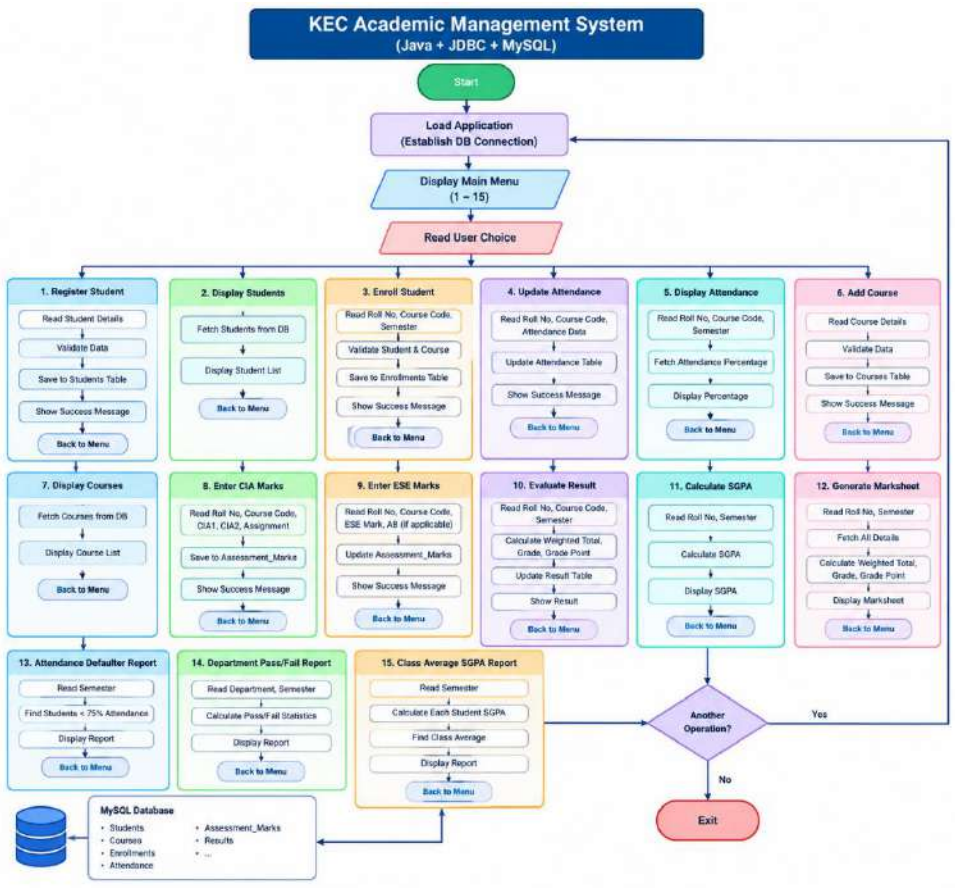
Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to

Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

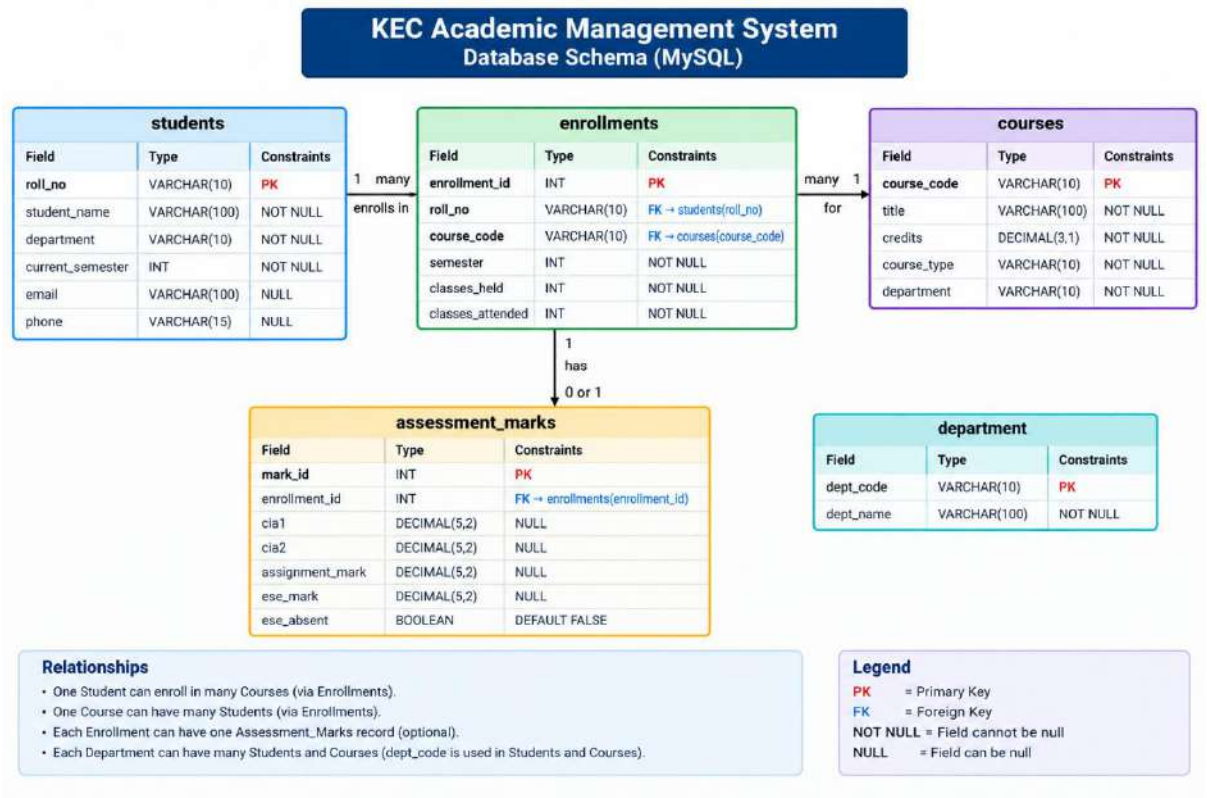
FLOW CHART:



PROCEDURE:

1. Install and configure Java JDK, VS Code, MySQL Server, and MySQL Connector/J.
2. Create a Java project named KEC-Academic-System and add the MySQL Connector/J library.
3. Create the database kec_academics in MySQL and define the required tables for students, courses, enrollments, and assessment marks.
4. Create DBConnection.java to establish a connection between Java and MySQL using JDBC.
5. Create StudentDAO.java to register and display student details.
6. Create AcademicDAO.java to manage courses, enrollment, attendance, marks, grades, and academic reports.
7. Create MainApp.java to display the menu and call the appropriate methods based on the user's choice.
8. Execute the SQL script and verify the database connection.
9. Run MainApp.java and test student registration, course enrollment, attendance updates, and marks entry.
10. Verify the stored records and academic results to ensure the system functions correctly.

DATABASE SCHEMA:



SOURCE CODE:

1. DBConnection.java

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBConnection {

    private static final String URL =
            "jdbc:mysql://localhost:3306/kec_academics";

    private static final String USER = "root";
    private static final String PASSWORD = "YOUR_MYSQL_PASSWORD";

    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(URL, USER, PASSWORD);
    }
}
```

2. StudentDAO.java

```
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;

public class StudentDAO {

    public void addStudent(String rollNo, String name,
                           String department, int semester) {

        String sql = "INSERT INTO students "
```

```
            + "(roll_no, student_name, department, current_semester) "
            + "VALUES (?, ?, ?, ?)";

    try (Connection con = DBConnection.getConnection();
         PreparedStatement ps = con.prepareStatement(sql)) {

        ps.setString(1, rollNo);
        ps.setString(2, name);
        ps.setString(3, department);
        ps.setInt(4, semester);

        ps.executeUpdate();
        System.out.println("Student registered successfully.");

    } catch (SQLException e) {
        System.out.println("Registration error: " + e.getMessage());
    }
}

public void displayStudents() {

    String sql = "SELECT roll_no, student_name, department, "
            + "current_semester FROM students ORDER BY roll_no";

    try (Connection con = DBConnection.getConnection();
         PreparedStatement ps = con.prepareStatement(sql);
         ResultSet rs = ps.executeQuery()) {

        System.out.println("\n===== STUDENT LIST =====");

        while (rs.next()) {
            System.out.println(
```

```
                rs.getString("roll_no") + " | "
                + rs.getString("student_name") + " | "
                + rs.getString("department") + " | Semester "
                + rs.getInt("current_semester"));
        }

    } catch (SQLException e) {
        System.out.println("Display error: " + e.getMessage());
    }
  }
}
```

3.MainApp.java

```
import java.util.Scanner;

public class MainApp {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      StudentDAO students = new StudentDAO();
      AcademicDAO academic = new AcademicDAO();

      while (true) {
         System.out.println("\n=== KEC ACADEMIC MANAGEMENT SYSTEM ===");
         System.out.println("1. Register Student");
         System.out.println("2. Display Students");
         System.out.println("3. Enroll Student");
         System.out.println("4. Update Attendance");
         System.out.println("5. Display Attendance");
         System.out.println("6. Add Course");
```

```
System.out.println("7. Display Courses");
System.out.println("8. Enter CIA Marks");
System.out.println("9. Enter ESE Marks");
System.out.println("10. Evaluate Results");
System.out.println("11. Calculate SGPA");
System.out.println("12. Generate Marksheet");
System.out.println("13. Attendance Defaulters");
System.out.println("14. Department Pass/Fail Report");
System.out.println("15. Class Average SGPA");
System.out.println("0. Exit");
System.out.print("Enter choice: ");

try {
    int choice = Integer.parseInt(sc.nextLine());

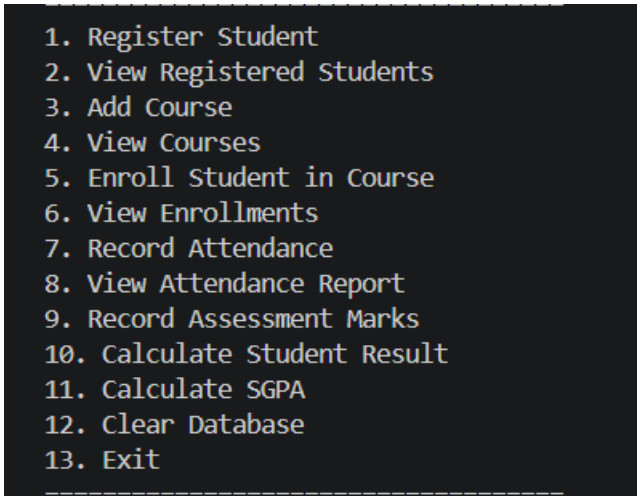
    switch (choice) {
        case 1:
            students.addStudent();
            break;
        case 2:
            students.displayStudents();
            break;
        case 3:
            academic.enrollStudent();
            break;
        case 4:
            academic.updateAttendance();
            break;
        case 5:
            academic.displayAttendance();
            break;
        case 6:
```

```
            academic.addCourse();
            break;
        case 7:
            academic.displayCourses();
            break;
        case 8:
            academic.enterMarks(true);
            break;
        case 9:
            academic.enterMarks(false);
            break;
        case 10:
            academic.evaluateStudent();
            break;
        case 11:
            academic.calculateSGPA();
            break;
        case 12:
            academic.generateMarksheet();
            break;
        case 13:
            academic.attendanceDefaulterReport();
            break;
        case 14:
            academic.departmentPassFailReport();
            break;
        case 15:
            academic.classAverageSGPAReport();
            break;
        case 0:
            System.out.println("Exiting system...");
            sc.close();
```

```
                return;
            default:
                System.out.println("Invalid choice!");
        }
      } catch (Exception e) {
        System.out.println("Error: " + e.getMessage());
      }
    }
  }
}
```

OUTPUT SCREENSHOTS:

MAIN MENU:



OUTPUT

1.STUDENT REGISTRATION:

```
==================================
        KEC ACADEMIC SYSTEM
==================================
1. Register Student
2. View Registered Students
3. Add Course
4. View Courses
5. Enroll Student in Course
6. View Enrollments
7. Record Attendance
8. View Attendance Report
9. Record Assessment Marks
10. Calculate Student Result
11. Calculate SGPA
12. Clear Database
13. Exit
==================================
Enter your choice (1-13): 1
Enter Roll Number: 24ITR128
Enter Student Name: SARAN
Enter Department: IT
Enter Current Semester: 3
Student registered successfully!
```

2. . View Registered Students:

```
        KEC ACADEMIC SYSTEM
==================================
1. Register Student
2. View Registered Students
3. Add Course
4. View Courses
5. Enroll Student in Course
6. View Enrollments
7. Record Attendance
8. View Attendance Report
9. Record Assessment Marks
10. Calculate Student Result
11. Calculate SGPA
12. Clear Database
13. Exit
==================================
Enter your choice (1-13): 2

===== REGISTERED STUDENTS =====
Roll Number: 24ITR128
Name: SARAN
Department: IT
Semester: 3
```

3. ADD COURSE:

```
==================================
        KEC ACADEMIC SYSTEM
==================================
1. Register Student
2. View Registered Students
3. Add Course
4. View Courses
5. Enroll Student in Course
6. View Enrollments
7. Record Attendance
8. View Attendance Report
9. Record Assessment Marks
10. Calculate Student Result
11. Calculate SGPA
12. Clear Database
13. Exit
==================================
Enter your choice (1-13): 3
Enter Course Code: 25ITT32
Enter Course Title: JAVA
Enter Credits: 4
Enter Course Type (THEORY/PRACTICAL/EMBEDDED): THEORY
Course added successfully!
```

4.VIEW COURSES:

```
        KEC ACADEMIC SYSTEM
==================================
1. Register Student
2. View Registered Students
3. Add Course
4. View Courses
5. Enroll Student in Course
6. View Enrollments
7. Record Attendance
8. View Attendance Report
9. Record Assessment Marks
10. Calculate Student Result
11. Calculate SGPA
12. Clear Database
13. Exit
==================================
Enter your choice (1-13): 4

===== COURSE CATALOG =====
Course Code: 25ITT32
Course Title: JAVA
Credits: 3
Type: THEORY
---------------------------
```

5. Enroll Student in Course:

```
        KEC ACADEMIC SYSTEM
==================================
1. Register Student
2. View Registered Students
3. Add Course
4. View Courses
5. Enroll Student in Course
6. View Enrollments
7. Record Attendance
8. View Attendance Report
9. Record Assessment Marks
10. Calculate Student Result
11. Calculate SGPA
12. Clear Database
13. Exit
==================================
Enter your choice (1-13): 5
Enter Student Roll Number: 24ITR128
Enter Course Code: 25ITT32
Enter Semester: 3
Student enrolled successfully!
```

6. View Enrollments:

```
==================================
        KEC ACADEMIC SYSTEM
==================================
1. Register Student
2. View Registered Students
3. Add Course
4. View Courses
5. Enroll Student in Course
6. View Enrollments
7. Record Attendance
8. View Attendance Report
9. Record Assessment Marks
10. Calculate Student Result
11. Calculate SGPA
12. Clear Database
13. Exit
==================================
Enter your choice (1-13): 6
Enter Student Roll Number: 24ITR128
Enter Course Code: 25ITT32
Enter Semester: 3
Roll Number: 24ITR128
Student: SARAN
Course Code: 25ITT32
Course: JAVA
Semester: 3

----------------------------------
```

7. Record Attendance:

```
==================================
        KEC ACADEMIC SYSTEM
==================================
1. Register Student
2. View Registered Students
3. Add Course
4. View Courses
5. Enroll Student in Course
6. View Enrollments
7. Record Attendance
8. View Attendance Report
9. Record Assessment Marks
10. Calculate Student Result
11. Calculate SGPA
12. Clear Database
13. Exit
==================================
Enter your choice (1-13): 7
Enter Student Roll Number: 24ITR128
Enter Course Code: 25ITT32
Enter Semester: 3
Enter Classes Held: 60
Enter Classes Attended: 56
Medical Condonation Approved? (Y/N): N
Attendance recorded successfully!
Attendance: 93.33% (56/60 classes)
Status: ESE Eligible
```

8. View Attendance Report:

```
==================================
        KEC ACADEMIC SYSTEM
==================================
1. Register Student
2. View Registered Students
3. Add Course
4. View Courses
5. Enroll Student in Course
6. View Enrollments
7. Record Attendance
8. View Attendance Report
9. Record Assessment Marks
10. Calculate Student Result
11. Calculate SGPA
12. Clear Database
13. Exit
==================================
Enter your choice (1-13): 8

===== ATTENDANCE REPORT =====

Roll Number: 24ITR128
Student: SARAN
Course: 25ITT32 - JAVA
Semester: 3
Attendance: 93.33% (56/60)
Status: ESE Eligible
------------------------------
```

9. Record Assessment Marks:

```
==================================
        KEC ACADEMIC SYSTEM
==================================
1. Register Student
2. View Registered Students
3. Add Course
4. View Courses
5. Enroll Student in Course
6. View Enrollments
7. Record Attendance
8. View Attendance Report
9. Record Assessment Marks
10. Calculate Student Result
11. Calculate SGPA
12. Clear Database
13. Exit
==================================
Enter your choice (1-13): 9

===== RECORD ASSESSMENT MARKS =====
Enter Student Roll Number: 24ITR128
Enter Course Code: 25ITT32
Enter Semester: 3
Enter CIA1 marks (0-100): 80
Enter CIA2 marks (0-100): 80
Enter Assignment marks (0-100): 87
Enter Quiz marks (0-100): 86
Enter Lab Evaluation marks (0-100, enter 0 if not applicable): 89
Was the student absent for ESE? (Y/N): N
Enter ESE marks (0-60): 55

Assessment marks saved successfully!
ESE Status: PRESENT
```

10.Calculate Student Result:

```
Enter your choice (1-13): 10

===== STUDENT RESULT CALCULATION =====
Enter Student Roll Number: 24ITR128
Enter Course Code: 25ITT32
Enter Semester: 3

========================================
          STUDENT RESULT
========================================
Roll Number : 24ITR128
Name        : SARAN
Department  : IT
Course Code : 25ITT32
Course Name : JAVA
Course Type : THEORY
Credits     : 3
Attendance  : 93.33%
Attendance Status: ESE Eligible
CIA Average : 84.40 / 100
CIA Weighted: 33.76
ESE Marks   : 55.00 / 60
ESE Weighted: 55.00
Final Total : 88.76 / 100
Grade       : A+
Grade Point : 9
Status      : PASS
========================================
```

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

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GITHUB:

LINK:	**https://github.com/ssarankarthik24-sudo/saranKarthik_javaJDBC**